



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS3102-1 – AUTOMATA THEORY AND COMPILER DESIGN

(For IS)

Max. Marks: 100

Note:

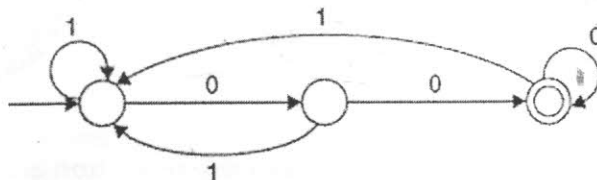
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A finite automaton recognizes**
A) Context-free language
B) Any language
C) Regular language
D) Context sensitive language
2. **The given DFA accepts the set of all strings over $\{0, 1\}$ that**



- A) begin either with 0 or 1
C) end with 00
- B) contain the substring 00
D) end with 0
3. Which of the following is true for the epsilon closure of a state q in a Non-deterministic Finite Automaton (NFA)?

A) It includes all states that can be reached from q via any input symbol.
C) It includes no states.

B) It includes only the state q itself.
D) It includes all states that can be reached from q via epsilon transitions.
4. In a compiler, the module that checks every character of the source text is called

A) The code generator
C) The lexical analyser

B) The code optimizer
D) The syntax analyser
5. Which of the following is used for grouping of characters into tokens?

A) scanner
C) code optimizer

B) parser
D) code generator
6. Which of the following is main operations in regular expression?

A) Union, Concatenation, Star closure
C) Concatenation, Intersection, Star closure

B) Union, Intersection, Star closure
D) Union, Intersection, Concatenation
7. A set of strings described by a rule is called

A) lexeme
C) pattern

B) token
D) grammar
8. Output file of Lex is _____, if the input file is Myfile

A) Myfile.e
C) Myfile.obj

B) Myfile.yy.c
D) Myfile.lex

9. Which of the following grammar has no two adjacent non-terminals?
 A) Regular Grammar B) Context Free Grammar
 C) Operator Grammar D) Ambiguous Grammar
10. The grammar $A \rightarrow AA \mid (A) \mid \epsilon$ is not suitable for predictive-parsing because the grammar is
 A) ambiguous B) left recursive
 C) right recursive D) an operator grammar
11. Given the CFG with the production rules $S \rightarrow AB$ and $A \rightarrow a, B \rightarrow b$, what string does it generate?
 A) bb B) ab
 C) aa D) aabb
12. What is a parse tree?
 A) A way to represent regular expressions. B) A graphical representation of the syntax of a string derived from a CFG.
 C) A type of finite state machine D) A linear representation of string characters
13. The difference between finite automata and push down automata is in _____.
 A) Finite control B) Reading head
 C) Stack D) Input tape
14. If a PDA accepts a string by empty stack, what must happen when processing the string?
 A) The stack must contain at least one symbol B) The input must end without any remaining symbols
 C) The stack must be empty after the string is processed D) The stack must have a specific symbol on top
15. Which of the following is not an action of the shift-reduce parser?
 A) shift B) reduce
 C) accept D) go
16. Consider the grammar
 $E \rightarrow E + n \mid E \times n \mid n$
 For a sentence $n + n \times n$, the handles in the right-sentential form of the reduction are
 A) $n, E + n$, and $E + E \times n$ B) $n, E + n$, and $E + n \times n$
 C) $n, E + n$, and $E \times n$ D) $n, n + n$, and $n + n \times n$
17. Which form of SDT uses both synthesized and inherited attributes with restriction of not taking values from right siblings?
 A) S-attributed SDT B) R-attributed SDT
 C) T-attributed SDT D) L-attributed SDT
18. _____ is a medium to provide semantics to the context-free grammar
 A) Attribute grammar B) Domain grammar
 C) Value grammar D) Object grammar
19. In a bottom-up evaluation of a syntax directed definition, inherited attributes can
 A) always be evaluated B) be evaluated only if definition is L attributed
 C) be evaluated only if definition is S attributed D) never be evaluated
20. Halting state of Turing machine are:
 A) Start and stop B) Start and reject
 C) Accept and reject D) Reject and allow

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Design a DFA to accept the language, $L = \{w \mid w \text{ is even and begins with } 01\}$. Write the extended transition function for i. 0101 ii. 1010	8	L*3	1	2

- b) Apply subset construction algorithm to convert the following NFA, Figure 1 (b) to DFA

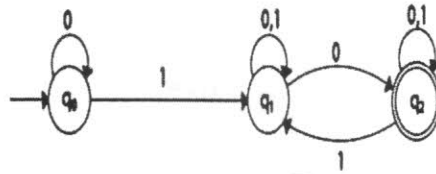


Figure 1(b)

8 L3 1 1

2. a) Minimize the following DFA, Figure 2(a)

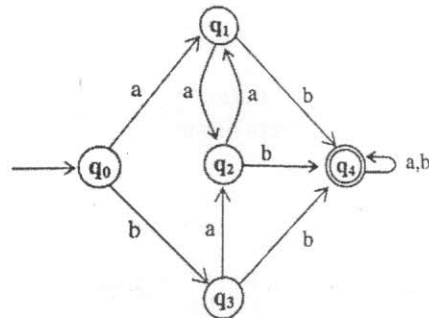


Figure 2 (a)

8 L3 1 2

- b) Draw a flowchart depicting various types of language processors and explain their roles and functionalities.
3. a) Explain input buffering and its importance in Lexical Analysis.
- b) Obtain regular expression for the following languages over $\{a,b\}$
- To accept strings of length two or more that begin and end with same symbol
 - string lengths are multiples of 3
 - Set of strings ending with either a or bb
 - strings with 3rd symbol from right is 'a' and 4th is 'b'

8 L2 1 1

8 L2 2 1

8 L3 2 2

Unit – II

4. a) Obtain context free grammar for the following language over $\{a,b\}$
- $L = \{ a^n b^{2n} : n, \geq 0 \}$ and find derivation tree for $w = aaabbbbbb$
 - Context free grammar to generate an arithmetic expression using the operators $+, -, *, /$ and $^$. Identifiers can start from any letters a, b, c and can be followed by zero or more symbols from $\{a,b,c\}$.
- b) Given the following grammar
- $A \rightarrow Ac \mid Sd \mid f$
- $S \rightarrow Aa \mid b$
- Perform the elimination of left recursion and apply left factoring to transform it into an equivalent grammar suitable for LL(1) parsing.
5. a) Check for ambiguity of following grammars
- $S \rightarrow AB \mid aaB$; $A \rightarrow a \mid Aa$; $B \rightarrow b$; $w = aab$
 - $SS \rightarrow S0S1S \mid S1S0S \mid \epsilon$, $W = 0101$
- b) Construct SLR Parsing table for the grammar and parse the string: "abbcde"
- $S \rightarrow aABe$
- $A \rightarrow Abc$
- $A \rightarrow b$
- $B \rightarrow d$

8 L3 3 2

8 L3 3 2

8 L3 3 2

8 L3 4 2

- | | | | | | |
|-------|---|---|----|---|---|
| 6. a) | Construct npda that accept the following language
$L = \{a^n b^m c^{n+m} : n, m \geq 0\}$. on $\Sigma = \{a, b, c\}$ | 8 | L3 | 4 | 2 |
| b) | Show that the following grammar is LR(1):
$S \rightarrow Aa \mid bAc \mid Bc \mid bBa$
$A \rightarrow d$
$B \rightarrow d$ | 8 | L3 | 3 | 2 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|---|
| 7. a) | Design a Turing Machine to accept the language $L = \{0^{2n}1^n \mid n \geq 0\}$ | 8 | L4 | 5 | 1 |
| b) | What are the key issues in the design of a code generator? Discuss how these issues impact the efficiency and correctness of the final generated code. | 8 | L2 | 5 | 2 |
| 8. a) | Write the Syntax Directed Definition for a simple desk calculator, using these rules build the annotated parse tree for the expression $1*2+3n$. | 8 | L3 | 5 | 1 |
| b) | Generate Three address code for the assignment
$A = b^*c + b^*c;$ | 8 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
